# Digital Tools

## Module Overview

- **Module:** EDS-M1 - Digital Tools
- **Course:** Essential Digital Skills (EDS)
- **Audience:** Early Career Researchers (PhD, Postdoc)

- **Duration:** ~40 minutes directed teaching (+4 exercises)

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## Learning Outcomes

- Navigate core UoN digital ecosystem
- Apply digital best practices in collaboration & file organisation
- Manage your research identity
- Identify essential practices for secure research

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## Questions

1. What basic data tools I need to do research at UoN?
2. How to I maintain a consistent online researcher identity?
3. What digital tools may I need to support my research aims?

### Structure & Agenda

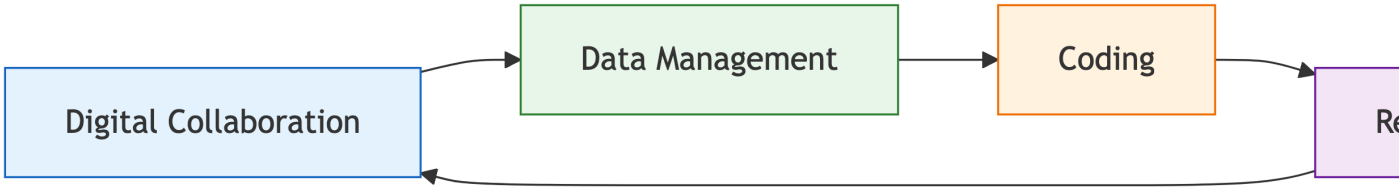
1. Introduction to the Digital Skills Programme (~10+10 min)
2. Institutional Systems for Digital Research (~10+10 min)
3. Research Identity and Profiles (~10+10 min)
4. Digital Security & Good Practice (~10+10min)

   4 Activities spaced every 10 minutes.

## Introduction to the Digital Skills Programme

### What is DS?

The **Digital Skills** programme equips researchers with the competencies required for modern, data-driven scholarship. it focuses on four interconnected themes:




*Digital collaboration, data management, coding, and reproducibility form the foundation of digital research practice.*

### Beyond Tools

The programme is not simply about tools:

It is about **connecting researchers** with the people who work with data, including **data stewards**, **software engineers**, and **information specialists**. These collaborations enable better use of data, improve the quality of research outputs, and support a more open and reproducible research culture.

*Digital skills grow through collaboration between researchers and data professionals.*

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### Connecting Researchers and Data Professionals

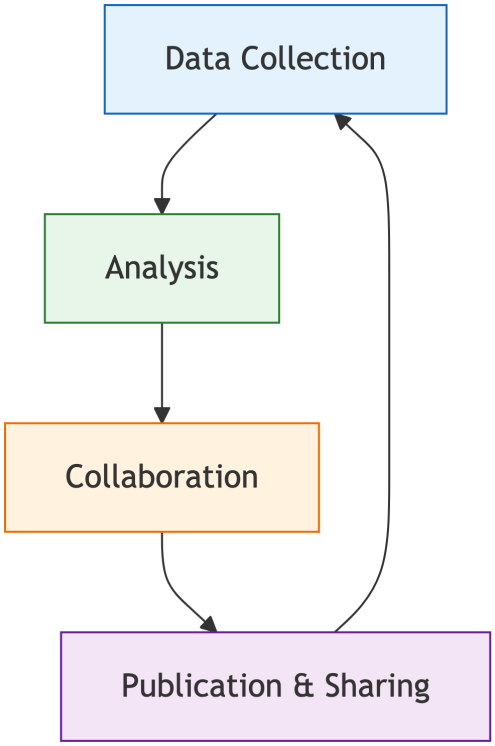
Modern research relies on shared understanding between disciplinary experts and technical specialists.

| Benefit | Description |
|---|---|
| **Quality** | Consistent data standards and metadata practices |
| **Reproducibility** | Shared methods and transparent workflows |
| **Efficiency** | Collaborative problem-solving across domains |
| **Insight** | Combining technical and disciplinary perspectives |

*Bridging communities builds stronger, more reproducible research.*

### Why Digital Skills Matter

Digital skills enable collaboration and efficiency across the research lifecycle.




*Digital literacy underpins the entire cycle of modern research.*

### Addressing Common Challenges

This course is about addressing the following challanges at a unversity level.

| Challenge | Digital Skills Response |
|---|---|
| Disconnected research silos | Shared tools and workflows (Teams, GitHub, RIS) |
| Data mismanagement | Version control, structured storage, FAIR principles |
| Limited collaboration | Integrated communication and documentation tools |
| Poor reproducibility | Use of open standards, coding literacy, and transparent sharing |

*Digital skills convert research challenges into structured improvements.*

## Collaboration Leads to Better Research

Collaboration is at the heart of the DS&DSP. By connecting researchers, data professionals, and support specialists, the programme builds a **community of digital practice** that strengthens the quality, openness, and impact of research across disciplines.

> "Collaboration is not an add-on—it is the mechanism through which good research becomes great research."

## Programme Structure

The programme offers **three tiers of learning**, from foundational to applied and specialist skills.

| Level | Focus | Example Modules |
|---|---|---|
| **Foundation** | Core digital research competencies | Essential Digital Skills |
| **Intermediate** | Theory and collaboration in data science | Python, R, Software Engineering |
| **Advanced** | Applied analysis and discipline-specific practice | Bioinformatics, HPC, ML/AI |

*Each tier builds practical capability and supports research independence.*

### Courses

| Course | Description | Duration | Note |
|---|---|---|---|
| Essential Digital Skills – Foundation | Introduces the tools, practices, and principles needed to participate confidently in digital research. | 4× half days | |
| Methods for Data Science – Theoretical | Introduces coding, software, and collaboration practices needed for sustainable research. | 4× half days | |

| Course | Description | Duration | Note |
|---|---|---|---|
| Methods for Data Science – Applied Python | Provides practical experience in data analysis and interpretation using Python. | 2 days | Introductory and intermediate |
| Methods for Data Science – Applied R | Provides practical experience in data analysis and interpretation using R. | 2 days | Introductory and intermediate |
| Advanced | — | 1 day? | AI? ML? Bioinformatics? |

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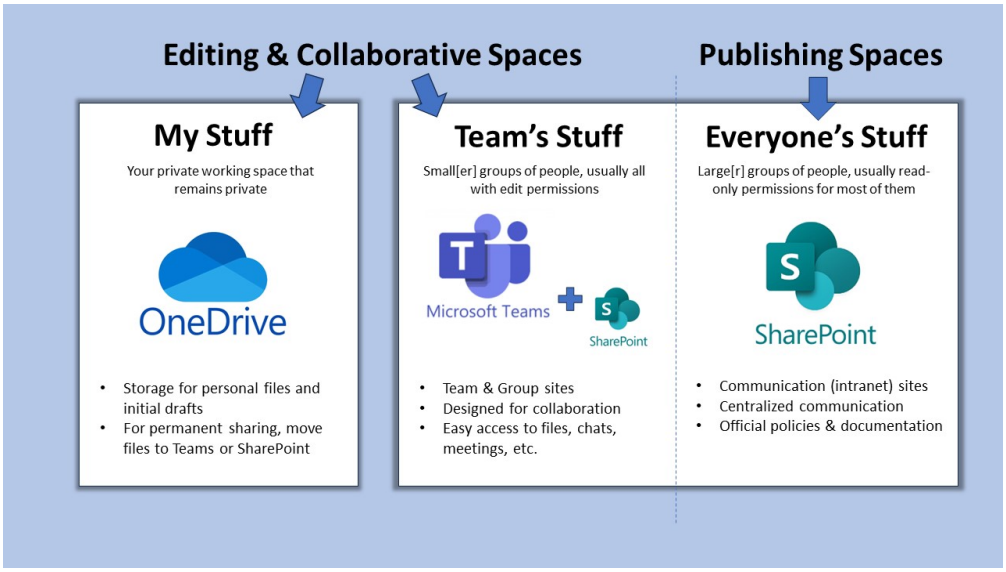
**Task 1: Digital Skills Reflection**

Let's start by reflecting on your current digital research skills.

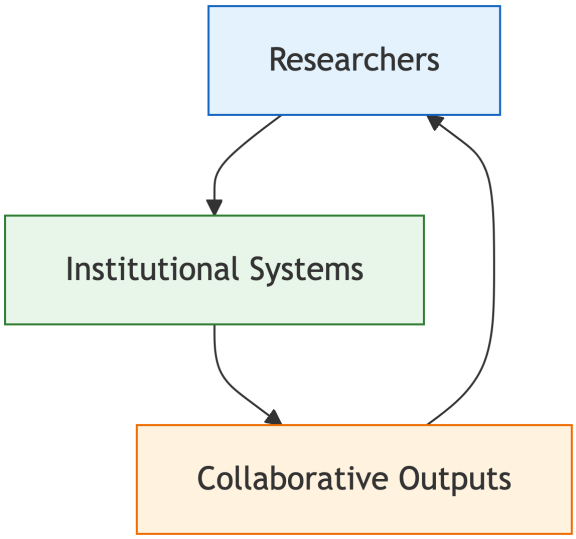
*What's digital skills or tool you want to develop this year?*

## Institutional Systems for Digital Research




### Why Institutional Systems Matter

Digital research depends on **connected systems** that enable collaboration, safeguard data, and streamline research workflows.




*Institutional systems form the backbone of a connected research ecosystem.*

## UoN's Digital Ecosystem

At the University of Nottingham, research operates within a **digitally integrated environment** that connects people, data, and administration.

| System | Primary Role |
|---|---|
| **Microsoft 365** | Collaboration and communication |
| **SharePoint** | Long-term data and document sharing |
| **UniCore** | Finance, HR, and procurement |
| **RIS (Worktribe)** | Research lifecycle, funding, outputs |

*Together these platforms enable secure, collaborative, and transparent research practice.*

## Microsoft 365: A Platform for Collaboration

Microsoft 365 supports daily research activity—from communication to data management—within a single, integrated workspace.

| Tool | Function |
|---|---|
| **Word & Excel** | Writing, data capture, and co-editing |
| **OneDrive** | Personal cloud storage with versioning |
| **Teams** | Communication, meetings, shared workspaces |
| **SharePoint** | Institutional document repositories |

*Microsoft 365 is the shared workspace for modern research collaboration.*

## Collaboration in Practice

Microsoft 365 enables seamless teamwork across roles and locations.

- Co-edit documents with real-time version tracking
- Use Teams for context-based discussion and meetings
- Store project materials in structured SharePoint libraries
- Sync files through OneDrive for offline access

*Integrated tools make collaborative research smooth, secure, and efficient.*

### File Collaboration Best Practice

- Use **Teams or SharePoint** for shared project files
- Enable **autosave and version history** in Word or Excel
- Maintain **logical folder structures** by project phase
- Apply **permissions and access control** with care

*Good file management is the foundation of digital reproducibility.*

### UniCore and RIS: Institutional Systems

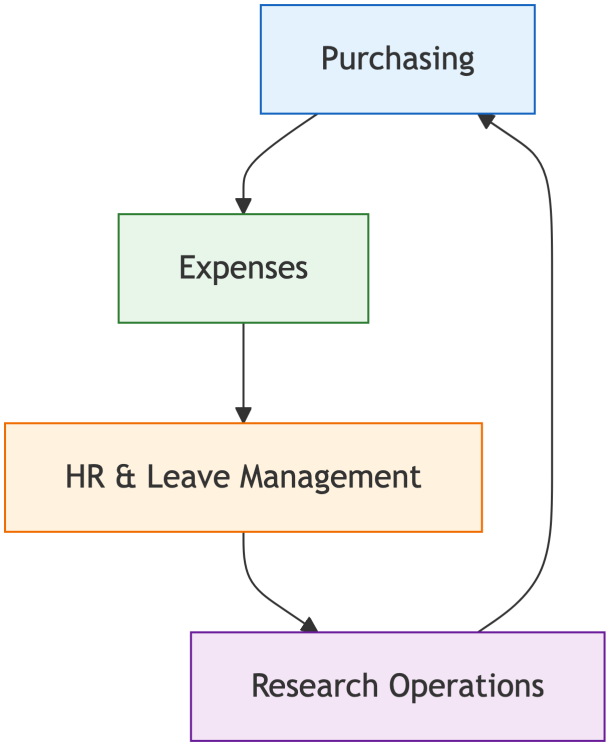
UniCore and RIS link the operational and administrative sides of research.

| System | Purpose |
|---|---|
| **UniCore** | Finance, HR, procurement, and resource management |
| **RIS (Worktribe)** | Project proposals, outputs, and compliance tracking |

*These systems ensure research activity is visible, auditable, and supported.*

### UniCore Overview

**UniCore** is UoN's enterprise system for managing finance, HR, and procurement.




*UniCore underpins the financial and operational side of research.*

### RIS (Research Information System)

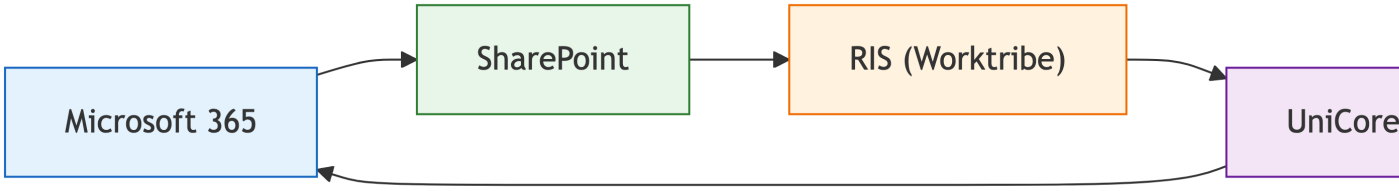
**RIS (Worktribe)** records, tracks, and reports the full research lifecycle.

| Feature | Purpose |
|---|---|
| **Grant Management** | Develop and cost funding proposals |
| **Outputs Tracking** | Record publications and datasets |
| **Impact Recording** | Capture engagement and REF evidence |
| **ORCID Integration** | Synchronise researcher identity and outputs |

*RIS ensures research activity is captured and connected institutionally.*

### Integrated Research Systems

The strength of UoN's infrastructure lies in **interoperability** — connecting research activity, collaboration, and administration.




*Connectivity between systems ensures seamless movement from idea to output.*

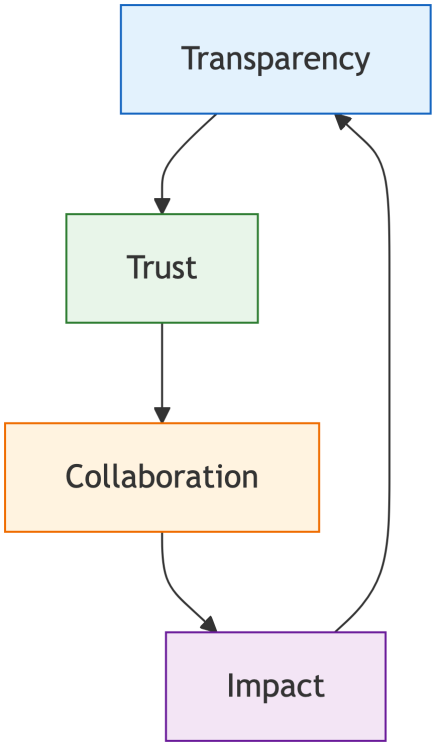
### System Use in Context

| Research Task | Recommended System | Collaboration Benefit |
|---|---|---|
| Co-author a paper | OneDrive / Word | Real-time editing and version control |
| Submit a funding bid | RIS | Shared proposal development and audit trail |
| Claim expenses | UniCore | Secure and traceable reimbursement |
| Share datasets | SharePoint / Teams | Controlled access and persistence |
| Record outputs | RIS + ORCID | Automatic institutional linking |

*Each platform serves a distinct role in the research ecosystem.*

### The Broader Vision: Connected Research

By using institutional systems effectively teams build **trust and accountability**, data remains **secure, reusable, and open** and researchers can spend more time working on **innovation and discovery**




*Digital infrastructure is the connective tissue of modern research.*

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### Task 2: Match Research Tasks to Tools

#### Activity

In groups of 4-5, discuss and list the digital tools you would use for each of the following:

1. Co-write a manuscript
2. Submit a funding bid
3. Order equipment
4. Share large datasets with your team

#### Cohort Answers

### Reflection

- Share your answers with the plenary
- Reflect on whether you've used any of these systems before
- Identify any gaps in your familiarity that you want to address

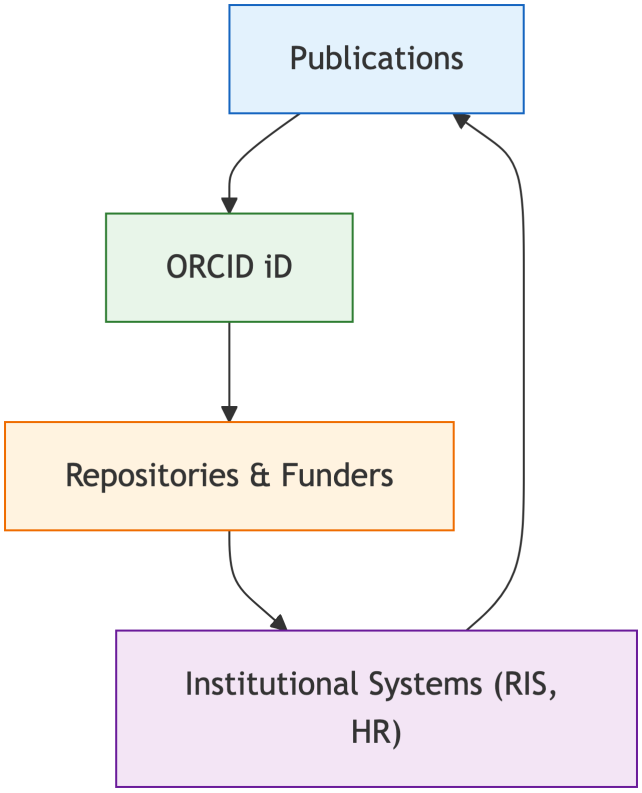
### Answers

- **Microsoft Word -> Sharepoint + Teams** are ideal for collaborative document editing
- **RIS (Worktribe)** is used for grant applications and research output tracking
- **UniCore** handles finance, procurement, HR and expenses
- **Microsoft Teams + SharePoint** is best for long-term group file access and collaboration

## Research Identity & Profiles

### Why Research Identity Matters

Your **research identity** defines how your work is discovered, attributed, and connected across the digital research ecosystem.




*A coherent digital identity ensures your contributions are visible, citable, and connected.*

### The Importance of Consistency

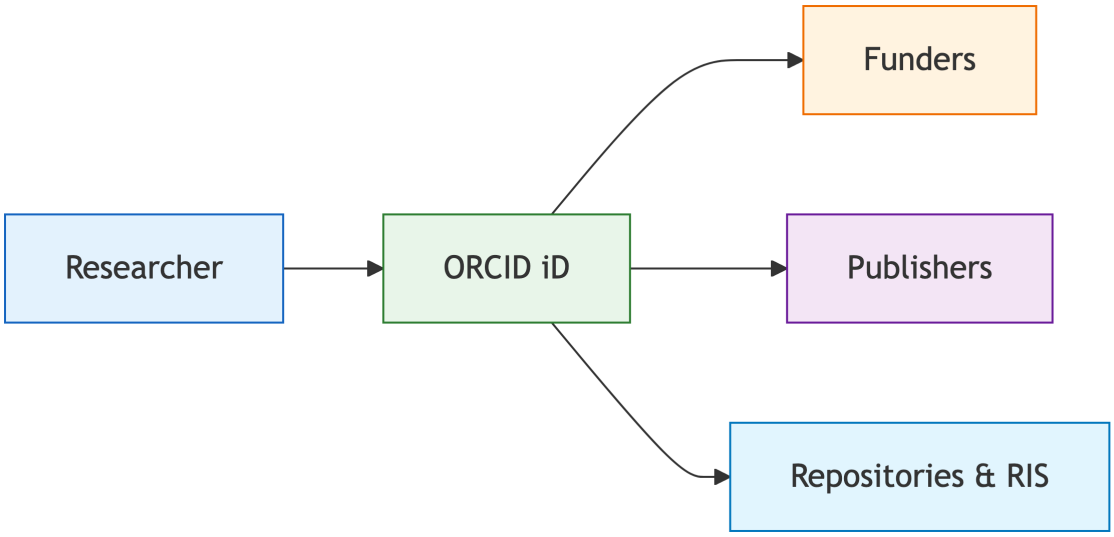
A consistent identity links your outputs, affiliations, and roles automatically across systems. This guarantees proper attribution and supports efficient collaboration.

- Ensures **discoverability** across publishers and repositories
- Prevents **duplication or misattribution** of outputs
- Facilitates **integration** with institutional systems like RIS

*A consistent research identity builds long-term recognition and credibility.*

## ORCID: A Persistent Digital Identifier

**ORCID (Open Researcher and Contributor ID)** is the global standard for uniquely identifying researchers across institutions and disciplines.




*Your ORCID iD is your digital fingerprint across the global research ecosystem.*

## Why ORCID is Essential

- **Universal interoperability:** used by funders, publishers, and institutions
- **Automatic updates:** Crossref and DataCite can sync new outputs
- **Portability:** your iD remains active throughout your career

*ORCID connects your research identity beyond institutional boundaries.*

## RIS (Research Information System) at UoN

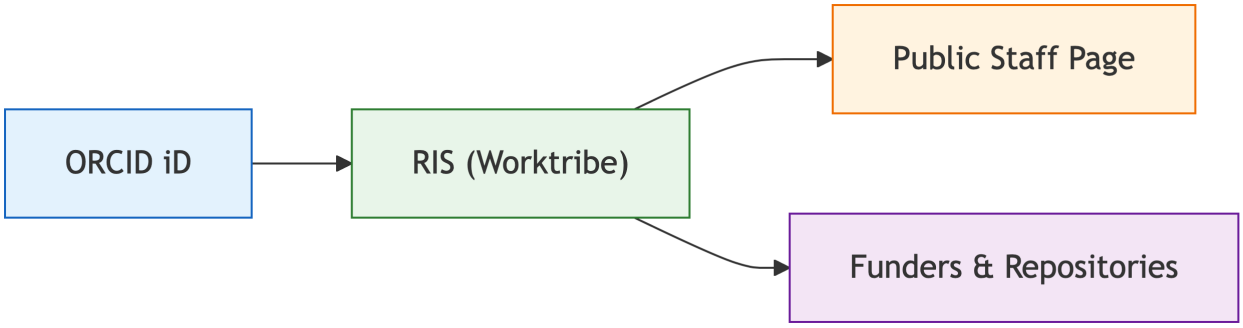
At UoN, **RIS (Worktribe)** complements your ORCID profile by linking research activity to institutional records.

| Feature | Function |
|---|---|
| **Grant Management** | Tracks proposals, approvals, and costings |
| **Outputs Dashboard** | Records publications, datasets, and impacts |

| Feature | Function |
|---|---|
| **Compliance** | Supports REF and Open Access requirements |
| **Integration** | Syncs with ORCID and external repositories |

*RIS ensures your institutional identity mirrors your global research presence.*

## ORCID–RIS Integration




*Connecting ORCID to RIS automates data flow and strengthens institutional visibility.*

## Scholarly Profiles Across Systems

Different platforms amplify your professional presence.
When synchronised with ORCID, they create a distributed network of your scholarly identity.

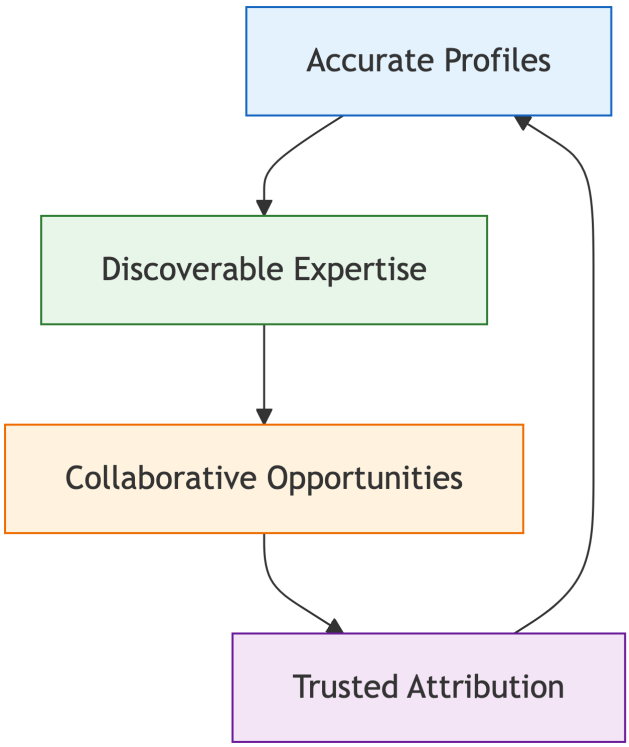
| Platform | Purpose | Notes |
|---|---|---|
| **Scopus Author ID** | Tracks citations, co-authorship, and affiliations | Merge duplicates for accuracy |
| **Google Scholar** | Monitors visibility and citation trends | Manual curation may be required |
| **ResearchGate / Academia.edu** | Social exchange platforms | Use responsibly; metrics are non-standard |

*Prioritise verified systems like ORCID, Scopus, and RIS for credible visibility.*

### Managing Your Research Profiles

- Make your **ORCID profile public** for discoverability
- **Merge duplicate Scopus IDs** to ensure accuracy
- **Link ORCID to RIS** for automatic synchronisation
- Use consistent **name and affiliation formats**
- Avoid unverified or **proprietary metrics**
- **Audit profiles regularly** for correctness

> *Managing your identity actively ensures your research remains visible and verifiable.*

### Research Identity as Collaboration Infrastructure

A connected identity supports collaboration, attribution, and trust across teams.